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Investigating carbon footprint and carbon reduction potential using a cradle-to-cradle LCA approach on lithium-ion batteries for electric vehicles in China

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ABSTRACT

Lithium-ion battery (LIB) is one of the core components of electric vehicles (EVs), and its ecological impacts are significant for the sustainable development of EVs. In this study, the carbon footprint of LIBs produced in China is investigated using a cradle-to-cradle life-cycle assessment approach. The results can be summarized as follows: (1) The carbon emission from battery production is 91.21 kg CO_2-eq/kWh, in which the cathode production and battery assembly process are the main sources of carbon emissions; (2) The carbon emission during the battery use phase under China's electricity mix which is dominated by thermal power in 2020 is 154.1 kg CO_2-eq/kWh; (3) Hydrometallurgy is the battery recycling method with the lowest carbon emission, and direct physical recycling method has attractive carbon reduction potential. (4) The carbon emission of battery remanufacturing through recycled materials is 51.8% lower than that of battery production with raw materials. Under the future electricity mix, the carbon emissions of battery production in 2050 and 2060 will be reduced by 75% and 84.9%, respectively. The short-term carbon reduction measure is battery recycling, and the long-term carbon reduction measure is the greening of electricity.

1. Introduction

Global climate change is posing a significant threat to human society. As a major greenhouse gas (GHG) associated with global warming, carbon dioxide will bring a series of problems, such as ice-sheet melt, sea-level rise, and ecological environment damage (Golledge et al., 2019). The Paris Agreement has set a temperature control target of 1.5 °C, and more than 190 countries have submitted "intended nationally determined contributions", indicating that they will achieve the economy-wide goal of reducing GHG emissions (Tanaka et al., 2021). Transportation is the second-largest GHG emission source in the world (Elgowainy et al., 2018), which has attracted widespread attention in carbon emission reduction. Vigorously promoting electric vehicles is an effective path to reducing GHG emissions and energy consumption, which has been adopted by governments around the world (Knobloch et al., 2020; Lai et al., 2021b). The rapid development of EVs has brought about large-scale applications of power batteries (Lai et al., 2021e). Lithium-ion batteries are the first choice for EVs because of their high energy density, high working voltage, low self-discharge rate, long life cycle, and almost zero memory effect (Lai, X. et al., 2022; Lai et al., 2021d). It is estimated that the global demand for LIBs will reach 9300 GWh by 2030 (Yuan et al., 2021), and the installed capacity of power LIBs has exceeded that of the electronic consumer market driven by the rapid development of EVs (Li et al., 2021). China's power battery production and sales ranked first in the world in 2014, which reached 220 GWh in 2021 with a year-on-year increase of 165%. Fig. 1 shows that the annual installed capacity of power batteries in China will reach about 1.5 TWh by 2030 (Union). However, the rapid growth of power battery production and use brings the problem of resource shortage and energy consumption, resulting in direct or indirect GHG emissions. Compared with the developed countries, the pressure on low-carbon energy in China is more significant. China is the world's largest producer and user of LIBs. Therefore, it is urgent to accurately calculate the carbon emission of LIBs in the whole life cycle in China under various electricity

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mixes to tap the potential of carbon emission reduction.

Life cycle assessment (LCA) is a method to assess the environmental burden in the life cycle of products, processes, or activities (Silvestri et al., 2020). Currently, LCA is widely used to evaluate environmental pollution and GHG emissions of various battery technologies, and they can be summarized as follows: (1) Comparing carbon emissions and environmental indicators between LIBs and other batteries. The results show that solid-state batteries produce lower carbon emissions than LIBs (Smith et al., 2020). Compared with the Ni–H battery and solar cell, the carbon emission of the raw material assembly process of LIBs is the lowest, and it has better environmental friendliness (Liang et al., 2017). Moreover, almost all environmental impact indicators of LIBs are lower than those of lead-acid and Ni–Cd batteries (Hammond and Hazeldine, 2015; Yang et al., 2020). (2) Comparison of carbon emissions and environmental indicators of LIBs with different material systems. The previous study (Vandepaer et al., 2017) showed that the GHG emissions and ozone consumption of lithium metal polymer batteries in the production stage are greater than those of the traditional LIBs. The carbon emission of Lithium-iron phosphate (LFP) batteries is about 42.0–44.5% lower than that of LIBs with nickel-cobalt-manganese oxide (NCM), and the value of water footprint and ecological footprint of LFP is lower (Wu et al., 2019). Compared with the NCM battery, the LFP battery shown less environmental burden in the production phase but a more significant environmental burden in the use and transportation phases (Shu et al., 2021). (3) Carbon emissions and environmental impacts of LIBs at different phases. The early LCA studies mainly focused on evaluating battery production and use stages due to the lack of recycling process data (Liang et al., 2017; Vandepaer et al., 2017; Wang, C. et al., 2017). The studies found that the carbon emission and environmental impact of the battery production stage dominate the whole life cycle of LIBs (Porzio and Scown, 2021; Vandepaer et al., 2017; Wang, C. et al., 2017). The environmental impact of the battery in the usage stage is mainly related to the electricity mix and the performance of the battery, such as the aging rate, charge, and discharge efficiency (Lybbert et al., 2021; Vandepaer et al., 2017).

With the rapid growth of EVs and the attention to global warming, the carbon emissions of LIBs in the whole life cycle are becoming the focus of governments, enterprises, and research institutions worldwide. However, the existing study results are quite different, and the carbon emission in the production phase ranges from 36.9 to 313 kg CO_2-eq/kWh, with an average of 110 kg CO_2-eq/kWh (Peters et al., 2017; Temporelli et al., 2020). The main reasons for the differences are that each study may be based on different battery life cycle inventory data, and some studies did not trace the upstream process data of the battery materials. In addition, few studies focus on the LCA study on battery remanufacturing, which refers to reusing recycled materials from waste LIBs in battery manufacturing. Note that the carbon footprint and carbon emissions are two indicators that are often mentioned (Lai et al., 2022), and the former has a larger scope than the latter. Generally, the carbon footprint is defined as the total amount of GHG emissions that come from the production, use, and end-of-life of a product or service, and the carbon emission is the GHG emission of a product or service at a particular stage or time. Because LIBs have a long-life cycle with different stages, it is necessary to investigate the carbon footprint of LIBs in this study and the potential carbon reduction measures proposed.

The current battery LCA studies mainly focus on the raw material extraction to the disposal (cradle-to-grave) or the battery production stage (cradle-to-gate) (Kim et al., 2016; Peters et al., 2017; Sanfélix et al., 2015), which are the typical open-loop LCA methods. With the development of battery material recycling and remanufacturing, the cradle-to-cradle method is attracting more and more attention. Specifically, the cradle-to-cradle method is a new full-lifecycle LCA method considering reusing waste battery materials and battery remanufacture. In this study, a cradle-to-cradle LCA method is developed to investigate the carbon footprint of LIBs in China under the reality and future scenarios, and some valuable carbon reduction measures are given. The study object is the $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ (NCM811) battery, which is widely used in EVs. The potential contributions of this paper are summarized as follows:

(1) The cradle-to-cradle LCA is conducted on the NCM811 battery produced and used in China.
(2) The carbon footprint in the full lifespan of LIBs, including battery production considering raw materials, use, recycling, and remanufacturing stages, are investigated in detail, and some valuable carbon reduction measures are suggested.
(3) The carbon emissions of three typical battery recycling methods are calculated and compared, followed by the carbon emission of the battery remanufacturing based on recycled materials and that of the battery production based on raw materials are compared. The results can quantitatively describe the carbon emission benefits of various battery recycling methods.
(4) The carbon emissions of battery production under the future electricity mixes are calculated, and the goal of carbon neutralization during battery production is prospected.

The study results are beneficial to determining the key factors affecting carbon footprint in the full lifespan of LIBs, and form some effective carbon emission reduction strategies facing China's carbon neutralization strategy. The remainder of this paper is organized as follows. Section 2 introduces the LCA approach. In Section 3, the carbon footprint of LIBs from cradle to cradle is calculated and discussed in rich scenarios, followed by some value carbon reduction measures are suggested. Finally, some conclusions are provided in Section 4.

2. Materials and methods

According to the definition of ISO 14044 standard, the basic structure of LCA mainly includes four parts, namely goal and scope definition, inventory analysis, impact assessment, and interpretation (Tolomeo et al., 2020). It is a technology and method used to assess the environmental impact of products in the whole life cycle.

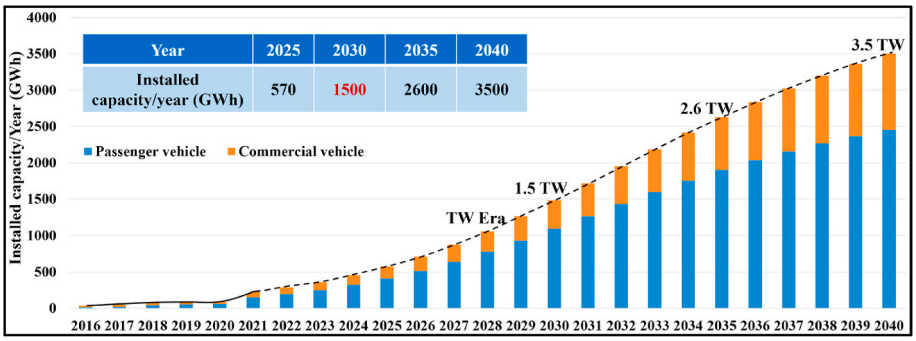


Fig. 1. Forecast of annual installed capacity of LIBs in China (Union).

2.1. Goal and scope

The study aims to quantify the carbon footprint of LIBs in the whole life cycle, assess the potential of carbon emission reduction in different stages, and investigate the carbon emissions in the production and use stages of LIBs for the carbon peak and carbon neutralization scenarios. The study results provide a valuable reference for the whole LIB industry chain to achieve the peak of carbon emission in 2030 and carbon neutralization in 2060. In this study, the carbon footprint model of the NCM811 battery from cradle to cradle is constructed based on the Gabi software version 10.6, and the carbon emission and its distribution at each stage are calculated and discussed.

2.1.1. Functional unit

The functional unit is used to formally define the studied product according to its function and allows unified analysis of several stages according to a functionally equivalent basis (Vandepaer et al., 2017). In this study, an NCM 811 battery is taken as the study object because of its wide application in EVs. The functional unit is defined as 1 kWh of the NCM811 battery for an EV, which fully ensures the quantitative evaluation and comparability of carbon emissions at each stage in the full life cycle of LIBs for EVs. In this study, a battery pack with the capacity of 74 kWh in an EV with a service life of 8 years and an overall driving range of 120,000 km is taken as the research object during the battery use phase.

2.1.2. System boundary

From the perspective of the whole life cycle of the battery, the scope of the LCA for LIBs should be from the raw material extraction (cradle) to the disposal (grave), it is sometimes referred to as “cradle-to-grave”. However, most existing studies about the LCA of LIBs focus on the battery production stage, which is often called the “cradle-to-gate” (Kim et al., 2016). In recent years, with the continuous increase of retired LIBs (Lai et al., 2021a), the study on the environmental impacts of different battery recycling methods has increased (Lin et al., 2021; Rinne et al., 2021). The above research on the whole life cycle of LIBs is often referred to as “cradle to grave”. To the best of our knowledge, few studies have focused on the carbon footprint of batteries from cradle to cradle (i.e., considering the recycling and remanufacturing of battery materials). The system boundary of the study is shown in Fig. 2. A complete closed-loop “cradle-to-cradle” approach for the carbon footprint calculation of the NCM811 battery system is performed following the standard LCA procedures. The whole system mainly includes the extraction and processing of raw materials, the component manufacture of battery, battery production, the use phase of battery, and the recycling and disposal of battery. Noted that the carbon footprint of battery transportation and storage is beyond the scope of this study due to their complexity.

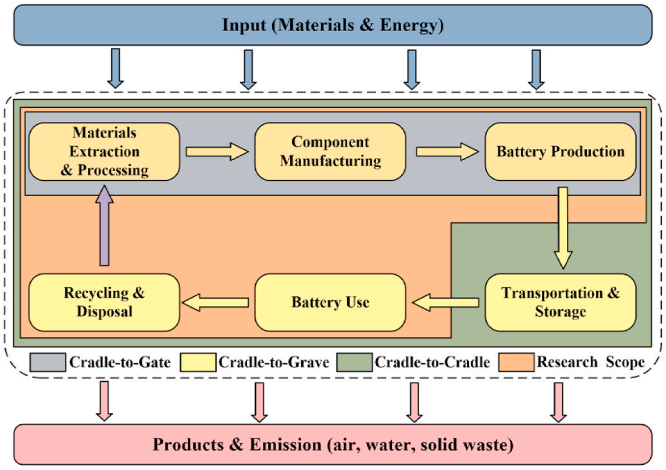


Fig. 2. The system boundary of the battery in the full life cycle.

2.2. Life cycle inventory analysis

Life cycle inventory (LCI) analysis is an essential step in creating input and output flow lists for a product system. The flows include inputs of water, energy, and raw materials, the outputs of products, and emissions to air, land, and water. Technically, LCI is the data collection portion of LCA, which can be obtained through experiments, enterprise research, literatures, and other ways, and the background data of upstream materials can be supplemented and corrected through LCI databases.

2.2.1. Inventory of battery production phase

The capacity of the investigated battery pack containing 188 cells is 74 kWh, and the weight and capacity of each cell are 1.74 kg and 116 Ah, respectively. The detailed information is listed in Table S1. Fig. 3 lists the material composition of a cell, including cathode, anode, separator, electrolyte, current collector, casing, and other materials. These data mainly come from our laboratory disassemble tests, and the LCI data of upstream material production is listed in Tables S3–S14. The test results reveal that the 1 kWh battery contains 559 g of cathode material, accounting for 34.43% of the total material inputs. The cathode is produced by the reaction of ternary precursor ($Ni_{0.8}Co_{0.1}Mn_{0.1}(OH)_2$) and lithium hydroxide using the high-temperature solid-phase method, co-precipitation method, and hydrothermal method (Gu et al., 2021). The ternary precursor is obtained from the mixed solution of manganese sulfate, nickel sulfate, and cobalt sulfate by adding liquid alkali (NaOH) and ammonia (Lai et al., 2022). Lithium hydroxide is the basic material for preparing ternary LIBs, which are mainly derived from lithium ore and lithium-containing brine (Wang, H. et al., 2017). The demand for anode material in batteries is second only to that of cathode material, which is 385 g/kWh, accounting for 22.13% of the total materials. The anode material is artificial graphite, which is obtained by high-temperature graphitization of natural graphite and pitch.

The battery separator in this study is 30 g/kWh, accounting for 2.87% of the total materials, and its raw materials mainly include polyolefins, polymer materials, and inorganic materials. The electrolyte (324 g/kWh) accounts for 18.62% of the input materials of the NCM811 battery, and it is prepared from lithium hexafluorophosphate ($LiPF_6$), ethylene carbonate (EC), and ethyl methyl carbonate (EMC) through low-temperature crystallization, filtration, and drying. The current collector of the NCM811 battery mainly includes two parts: the current collector aluminum foil of the cathode and the current collector copper foil of the anode. The cathode collector of the NCM811 battery uses aluminum (58 g/kWh), accounting for 3.33% and the anode collector uses copper (129 g/kWh), accounting for 7.41% of the input materials.

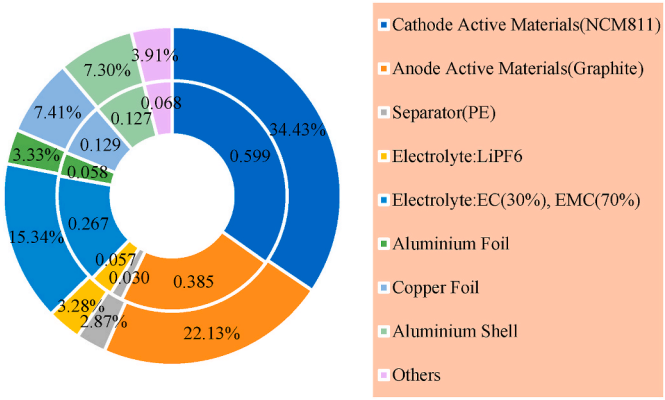


Fig. 3. The material composition of a commercial NCM811 battery.

In addition, shell production accounts for 7.30% of the total material input, and its primary component material is aluminum. Metal aluminum is produced from bauxite by alumina electrolysis. The production process needs to consume a lot of energy, and the pollution is also relatively severe. In China, pyrometallurgical smelting is the main method of producing metallic copper (Chen, J. et al., 2019), which has extensive energy consumption and produces many carbon emissions. Moreover, other parts account for 3.91% of the total material input, including NMP, PVDF, SBR, and other materials with less consumption in battery production.

The energy consumption in the battery production stage is measured according to the actual situation of the factory production line, mainly including the consumption of electricity and natural gas. As shown in Fig. 4, the battery production needs to go through a series of processes such as homogenization, coating, drying, rolling, die-cutting, packaging, liquid injection, and aging. Among them, the energy consumption in the drying/solvent recycling stage and dry room account for 38.15% and 35.77% of the total energy consumption of battery production, respectively. The main reason for the highest energy consumption of the dry room is that there are multiple process steps in the whole production process of the battery that needs to be carried out in a vacuum drying environment, and a continuous energy supply is required to keep the temperature constant.

2.2.2. Inventory of battery use phase

The LCI of the battery use phase mainly considers the service life and energy consumption of the NCM811 battery in EVs, and the service life is decided by its decay rate. The total driving distance of EVs in the battery service stage can be calculated as follows (Deng et al., 2018):

$$T = D_0\left(c - \frac{rc^2}{2}\right) \tag{1}$$

where D_0 is the initial driving distance per charge, r is the capacity decay rate, and c is the cycling number.

LIBs must be retired from EVs to ensure safety when their capacity decays to 70%–80% of the initial capacity (Lai et al., 2021c). Meanwhile, the capacity decay rate needs to be lower than 0.04% to keep the 200,000 km driving distance for a single battery pack (Wang et al., 2020). Electricity is the energy consumption during battery use, and the total energy consumption is related to many factors, such as battery efficiency, driving distance per charge, and battery charging efficiency. It can be expressed as follows (Deng et al., 2018):

$$E_{use} = \frac{E_{discharged}}{D\eta_{battery}\eta_{charger}} T \tag{2}$$

where $E_{discharged}$ is the discharged energy, T is the total driving distance in the battery service stage, D is the driving distance per charge, $\eta_{battery}$ is the battery efficiency, and $\eta_{charger}$ is the battery charge efficiency.

2.2.3. Inventory of battery recycling phase

At present, there are four main battery recycling methods, namely pyrometallurgy, hydrometallurgy, pyro- and hydrometallurgy, and direct physical recycling (Fan et al., 2020; Qian et al., 2022). Fig. 5 shows the pyrometallurgical technology route studied in this study. In pyrometallurgy, waste LIBs are melted in a high-temperature furnace. The valuable metals (nickel-cobalt-manganese) react with the reducing agents/additives to be reduced to cobalt-nickel-copper alloy products for recycling (Rajaeifar et al., 2021). In the process, lithium and aluminum are turned into slag and discarded, and the separator, electrolyte, and electrolyte are burned. It is a battery recycling technology route that has been widely commercialized.

The technical route of the combined pyro- and hydrometallurgy recycling batteries is shown in Fig. 5 (Doose et al., 2021). After the batteries are discharged, disassembled, and crushed, they are heat-treated at different temperatures to recover the high-value cobalt and nickel materials, and lithium is recovered through the further hydrometallurgical process. Such as the LCA of Umicore's VAL'EAS™ process is a typical pyro- and hydrometallurgy method (Fan et al., 2020). The hydrometallurgical process has the characteristics of high recycling efficiency, low energy consumption, simple reagent operation, and relatively mature technology. It's the mainstream process used by Chinese Enterprises. After the batteries are discharged, disassembled, and pulverized, the final product is obtained by hydrometallurgy through low-temperature leaching, purification, and separation. The leaching process includes alkali leaching and acid leaching, and the technical route of acid leaching is more mature and widely used in commercialization. However, strong acids and other chemical reagents in the acid leaching process will generate considerable secondary pollution, such as toxic gas and waste acid solution. The technical route of direct physical recycling can be described as follows: First, the waste batteries are decomposed into smaller particles by physical methods,

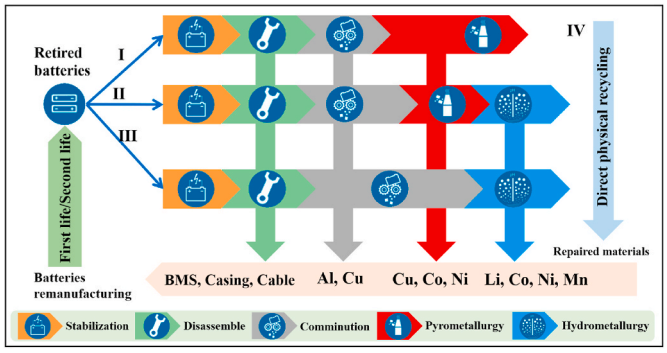


Fig. 5. Recycling routes of LIBs (Doose et al., 2021).

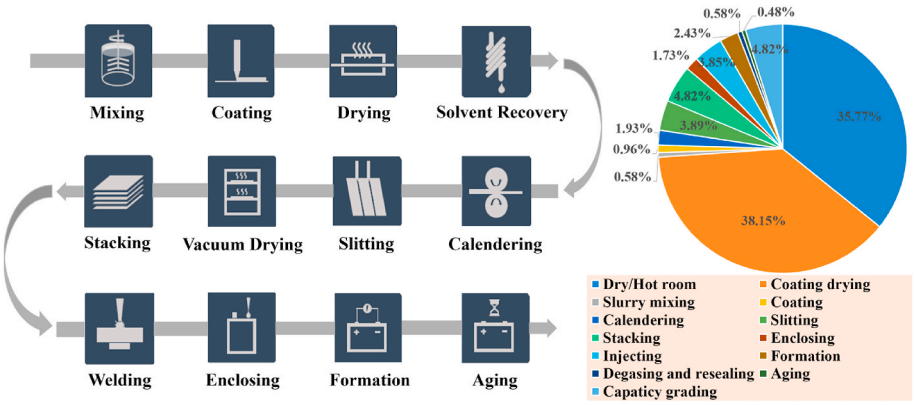


Fig. 4. Battery production process (Liu et al., 2021) and energy consumption distribution.

such as grinding or crushing. Then, these mixed particles are separated according to their physical characteristics, such as particle size, density difference, hydrophilicity, conductivity, and magnetism (Sommerville et al., 2020). The cathode collector aluminum foil and anode collector copper foil, separator, and anode active material graphite are recycled through the above process. The recycled cathode active material is mixed with nickel-rich material (such as $Ni_{0.83}Mn_{0.09}Co_{0.08}(OH)_2$ or $Ni(OH)_2$), $LiSO_4$, and LiOH to repair the ternary cathode material, and the LiOH is excess to facilitate complete re-lithiation of ternary cathode materials. The recycled materials can be used directly to produce new batteries, saving energy and resource consumption. In this study, the carbon emissions of pyrometallurgy, hydrometallurgy, and direct physical recycling methods are calculated and compared. The pyro- and hydrometallurgy recycling method is not within the scope of this study.

2.3. Data sources

Although NCM811 batteries have been widely used in EVs, few companies disclose their production, recycling process, and energy consumption. The data of each stage in the whole life cycle of the battery in this study are derived from battery disassembly, software, database, and literatures, as shown in Table 1. The detailed list of data can be found in Appendix A2.

3. Results and discussion

3.1. Carbon emission and carbon reduction potential during battery production

3.1.1. Carbon emission results

The carbon emissions of the battery production phase are calculated using Gabi 10.6 software, and the results are shown in Fig. 6. It can be observed that the total carbon emission during the battery production phase is 91.21 kg CO_2-eq/kWh, and the carbon emission from the production of cathode materials (47.15 kg CO_2-eq/kWh) accounts for the most significant proportion (51.69%) during the battery production. The primary contribution of carbon emissions of the cathode materials comes from upstream processes related to nickel, cobalt, lithium extraction, and beneficiation. As the lithium extraction technology has not yet reached the level of industrial production and geographical conditions, more than 70% of lithium relies on imports in China. Chile has the largest lithium brine resource globally, followed by Bolivia and Argentina, which account for nearly 80% of the world's brine reserves (Hu et al., 2021). The countries with rich brine have backward production equipment and sizeable primary energy consumption, which is the main carbon source. For countries rich in lithium mineral concentrates, such as Australia, extracting lithium from minerals produces higher carbon emissions than extracting lithium from brine (Kelly et al., 2021). Cobalt is the most valuable component of batteries, and two-thirds of the global cobalt supply is mined in the Democratic Republic of the Congo (Baars et al., 2020). Due to the instability of the national political situation and the backwardness of technology, many carbon emissions will also be generated in cobalt mining. Compared with cobalt, the global nickel reserves are widely distributed, in which laterite nickel accounts for 60% of the global terrestrial nickel resources and nickel sulfide ore accounts for 40%. The global nickel supply mainly comes from a few countries with advantages in nickel and coal resources, such as Indonesia. Nickel production in the above countries is characterized by high temperature, low yield, and high fossil energy consumption, which is also the main source of carbon emissions of cathode materials.

The carbon emission of battery assembly (28.11 kg CO_2-eq/kWh) is the second-largest contributor to battery production, mainly from electricity and natural gas. The carbon emission from electricity is 22.16 kg CO_2-eq/kWh, which is related to the power structure. Here, the average electricity mix of China in 2020 is considered: 67.9% of thermal power, 17.0% of hydropower, 6.0% of wind power, 3.5% of photovoltaic power generation, and 5.6% of other power generations (CEC, 2021). The other power generations include nuclear energy, geothermal energy, tidal energy, biomass energy, etc. As shown in Fig. 6, the carbon emission of other parts is 15.95 kg CO_2-eq/kWh, mainly including carbon emissions of the anode, separator, electrolyte, copper foil, aluminum foil, casing, and auxiliary materials. In this study, the casing accounts for the highest proportion in other parts, mainly because the casing used in the battery is aluminum. Upstream raw materials and energy of aluminum production, including electricity, natural gas, and limestone, will produce indirect carbon emissions, while the energy consumed in the process and chemical reactions will generate direct carbon emissions. The production of aluminum in China is dominated by electrolytic aluminum with high energy consumption and high emission, and the power sector is the primary source of carbon emission in the electrolytic aluminum industry chain, mainly due to the high proportion of thermal power in China's energy structure. The results in Fig. 6 show that the carbon emission from the anode accounts for 23.13% of that from other parts. The production of artificial graphite requires a high temperature, a long duration of carbonization, and a graphitization process, which will generate much carbon emissions. The carbon emission from the production of auxiliary materials accounts for 21.82% of that from other parts, including styrene-butadiene rubber (SBR), carboxymethyl cellulose (CMC), N-methyl-2-pyrrolidone (NMP), polyvinylidene fluoride (PVDF), deionized water, carbon nanotubes (CNT), and super P (SP), etc. The carbon emissions of copper and aluminum

Table 1
Data source of an NCM811 battery.

Phase	Inputs	Unit	Data source
Battery production	Cathode (NCM811)	g	(Shu et al., 2021), battery disassembly
	Anode (Artificial graphite)	g	(Zhao and You, 2019), GaBi 10.2, and battery disassembly
	Separator	g	(Silvestri et al., 2020), GaBi 10.2, and battery disassembly
	Electrolyte ($LiPF_6$)	g	(Yang and Guo, 2019)<comment message=The citation(s) 'Yang et al., 2019' has/have been changed to match the author name/date in the reference list. Please check here and in subsequent occurrences, and correct if necessary.></comment>, battery disassembly
	Copper foil	g	(Zhao and You, 2019), GaBi 10.2, battery disassembly
	Aluminum (foil, casing)	g	(Zhao and You, 2019), GaBi 10.2, and battery disassembly
	Binder (PVDF)	g	(Yuan et al., 2021), (Sun et al., 2020), and battery disassembly
	Electricity	kWh	Manufacturer
	Natural gas	MJ	Manufacturer
Use phase	electricity	kWh	, manufacturer
Recycling phase	Hydrometallurgy	-	(Rinne et al., 2021), (Lin et al., 2021), (Mohr et al., 2020)
	Pyrometallurgy	-	(Rajaeifar et al., 2021), (Mohr et al., 2020)
	Direct physical recycling	-	Qian et al. (2022)

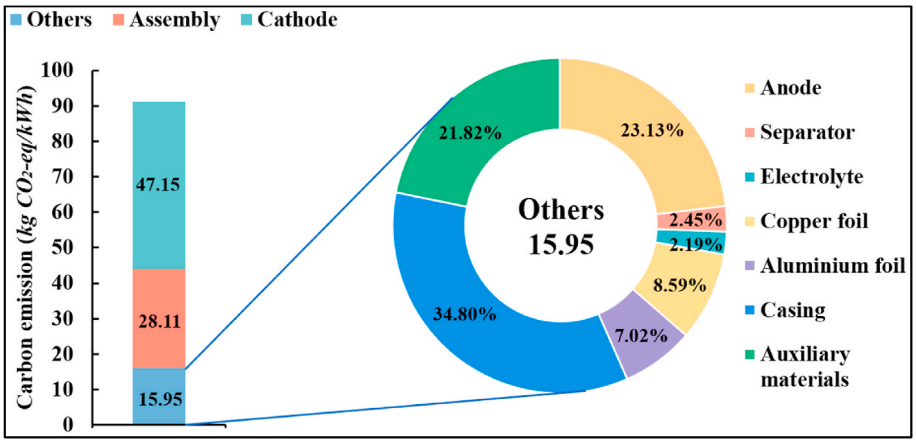


Fig. 6. Carbon emissions during battery production.

foils account for 8.59% and 7.02% of that from other parts, respectively. The upstream materials and energy consumption of aluminum foil production are consistent with aluminum casing, but there are more direct carbon emissions in the production process of aluminum foil because more processes and sophisticated equipment are required. Carbon emissions from copper production in China include indirect carbon emissions from copper production and direct carbon emissions from copper foil production. The production process of upstream copper and copper foil for LIBs is complex and long, the key production equipment is backward, and the annealing process is poor, which are all sources of high carbon emissions. The carbon emissions from electrolyte and separator account for 2.45% and 2.19% of that from the other parts, respectively, and the main source is carbon emissions from chemical reactions.

3.1.2. Comparison with the existing results

To verify the accuracy of the calculation results, the carbon emissions of battery production in this study are compared with those in existing studies, and the results are shown in Fig. 7. It can be observed that the carbon emissions of the production of different batteries have significant differences, ranging from 42 to 140 kg CO_2-eq/kWh, and the cathode, assembly, and other parts are different. In this study, the carbon emission of the NCM811 battery during production is 91.21 kg CO_2-eq/kWh. This is because the nickel content of the NCM811 battery is high, and the energy consumption of nickel extraction and processing is higher than that of cobalt and manganese (Crenna et al., 2021). The difference in carbon emissions during battery assembly is mainly due to the difference in the power grid structure. The carbon emissions in the assembly phase in this study are similar to those calculated by Dai et al. (Kelly et al., 2019), which are calculated based on China's electricity mix. However, our calculations are very different from those calculated by Ciez et al. (Ciez and Whitacre, 2019), which are calculated based on the electricity mix of the United States. The difference in carbon emission from other parts of the battery is mainly due to the different background data sources (Kelly et al., 2019; Sun et al., 2020), and the battery production in the early stage of commercialization used the relatively backward production equipment (Kim et al., 2016). As shown in Fig. 7, our calculation results are within a reasonable range, showing high reliability.

3.1.3. Carbon emission reduction potential in production

The carbon emission of battery production is affected by many variables in all links of the production chain, including the substitution of upstream materials, process improvement, improvement of energy efficiency, improvement of productivity, and reducing the carbon intensity in electricity. The impact of the future electricity mix shown in Table 2 on carbon emissions from battery production is analyzed in this study. The energy consumption used in battery production includes electricity and natural gas. Natural gas is used for drying after battery coating, rolling, and keeping the temperature of the processing workshop constant. The energy used in other battery production processes is electricity. Based on different electricity mixes in China, the carbon

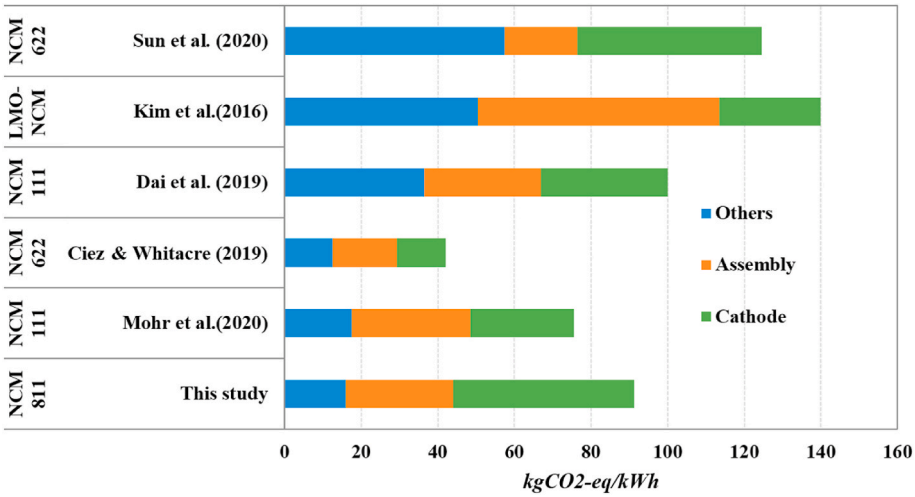


Fig. 7. Comparison of carbon emissions in the battery production phase.

Table 2
Future electricity mix in the next 30 years in China (CEC, 2021).

Item	2020	2030	2040	2050
Thermal power	67.9%	50.0%	32.0%	15.0%
Hydropower	17.0%	15.0%	12.0%	9.0%
Wind power	6.0%	17.0%	28.0%	38.5%
Photovoltaic power	3.5%	9.5%	15.5%	21.5%
Others	5.6%	8.5%	12.5%	16.0%

emissions during battery production can be calculated, and the results are shown in Fig. 8. It can be observed that the carbon emission during battery production under the average electricity mix in 2020 in China is 98.92 kg CO2-eq/kWh, which is 8.5% higher than that of actual battery production in 2020. The reason for this phenomenon is that China's electricity mix was dominated by thermal power in 2020. Thermal power generation is composed of coal power, oil power, and gas power, and the carbon emission of coal power is much higher than that of gas power. Therefore, the pure electricity mix dominated by coal-based power produces more carbon emissions than the mixed energy of natural gas and electricity during battery production.

China's electricity mix is becoming green and low-carbon. Table 2 lists the evolution of China's electricity mix in the next 30 years. Under the future electricity mix, the carbon emissions in the battery production are calculated, and the results are shown in Fig. 8. It can be seen that the carbon emission of battery production in 2030 is 80.45 kg CO_2-eq/kWh, which is 11.8% lower than that in 2020. In 2040 and 2050, the carbon emissions of battery production are 57.92 kg CO_2-eq/kWh and 22.98 kg CO_2-eq/kWh, respectively, which are reduced by 36.5% and 74.8% compared with 2020, respectively. The increase in the proportion of clean power cannot reduce carbon emissions indefinitely due to the carbon emissions of some chemical reaction processes, and a certain proportion of thermal power participating in peak shaving. It can be achieved net-zero carbon emission through carbon capture, utilization, and storage (CCUS) and direct air capture (DAC). From the above analysis, it can be concluded that low-carbon and green energy in electricity is the biggest factor to reduce carbon emissions during battery production in China. Combined with the new negative carbon technology, the goal of carbon neutrality in battery production can be achieved in 2060.

The improvement of production efficiency depends on various factors, such as production equipment, process, and economic benefits, which is a potential measure to reduce carbon emissions. According to our data survey in a leading battery manufacturer, the production efficiency of the battery batch production lines is about 30 ppm (pieces per minute) in China. In this study, the relationship between production efficiency and carbon emissions in the battery assembly is investigated. Considering the limitations of equipment and cost, it assumes that the battery production efficiency is increased by 5 ppm every year, and the maximum production efficiency is 50 ppm. As shown in Fig. 8(b), the carbon emission at the battery assembly stage is 28.11 kg CO_2-eq/kWh at the current battery production efficiency of 30 ppm. When the production efficiency is increased to 40 ppm, the carbon emission in the battery assembly stage is 21.08 kg CO_2-eq/kWh, which is 25% lower than that at 30 ppm. When the production efficiency is 50 ppm, the carbon emission in the battery assembly stage is 16.87 kg CO_2-eq/kWh, which is 40.0% lower than that at 30 ppm. With improved production efficiency, the energy consumption in the battery assembly stage is reduced, and the carbon emission is also reduced. It can be concluded that the carbon emission in the battery production phase can be reduced with the improvement of production efficiency to a certain extent. However, due to the limitation of production technology, production efficiency cannot be improved indefinitely, and the potential for carbon emission reduction will enter the plateau period.

3.2. Carbon emission and carbon reduction potential during battery use-phase

In this section, the carbon emission and the potential of carbon emission reduction in the use-phase of batteries under different electricity mixes are investigated. Electricity is the only energy used in the usage phase of EVs, which means that the carbon emissions in the use phase of batteries are mainly related to the carbon intensity in electricity in China. The carbon emissions in the battery usage stage under different electricity mix in China are calculated, and the results are shown in Fig. 9 (a). It can be observed that the carbon emission during battery usage is 154.1 kg CO_2-eq/kWh in 2020, and the carbon emission in the battery usage stage in 2030 will be reduced to 114.9 kg CO_2-eq/kWh. The latter is 25.4% lower than the former. Furthermore, the carbon emission in the battery use phase in 2040 and 2050 is 75.5 kg CO_2-eq/kWh and 38.3 kg CO_2-eq/kWh, respectively. It can be seen that the carbon emission in the battery use stage in 2050 will be 75.1% lower than that in 2020. The proportion of clean energy in China will be further increased by 2060. In this case, with the help of negative carbon technologies, Moreover, the use of negative carbon technologies such as CCUS and DAC can absorb the carbon emissions generated by the manufacturing of wind power and photoelectric infrastructure. It can be predicted that carbon neutralization in the battery use stage can be achieved in China in 2060.

China has a vast territory with different electricity mixes in different regions. Therefore, the carbon emissions of batteries used in different regions are different. In this study, the carbon emissions during battery usage in different regions of China are calculated, and the results are shown in Fig. 9 (b). The carbon emission of LIBs in East China in 2020 during the usage is 207.7 kg CO_2-eq/kWh, which is 1.35 times that of the national average. The reason for this phenomenon is that East China is still dominated by thermal power generation, such as in Shandong Province, the proportion of thermal power generation in 2020 accounts for 91.9% of the total power generation. Southwest China is rich in

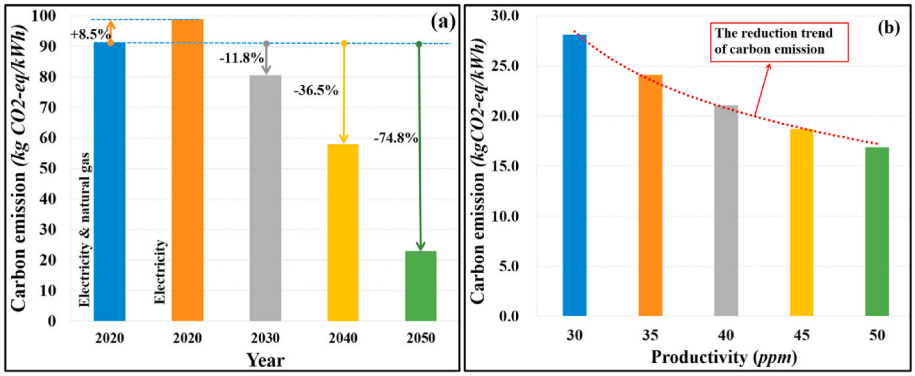


Fig. 8. Carbon emission reduction potential. (a) Different power situations of battery production; (b) Different productivity of battery assembly.

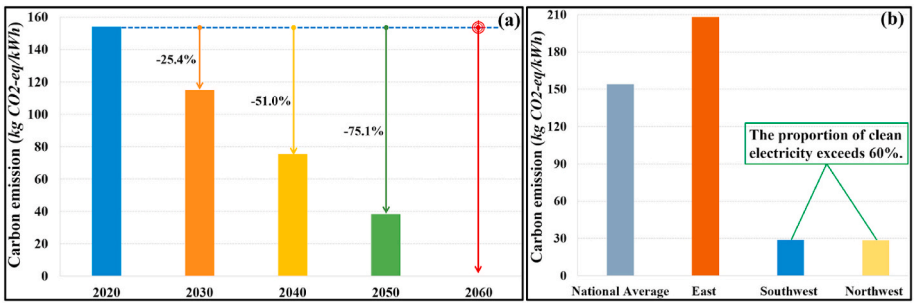


Fig. 9. Carbon emissions under different scenarios during the battery usage phase. (a) Future electricity mix; (b) Different regions of China.

hydropower resources, and the proportion of hydropower generation is high. For example, the proportion of hydropower generation in Yunnan Province reaches 80.1%. The carbon emission of LIBs during usage in 2020 in Southwest China is 28.7 kg CO_2-eq/kWh, which is 81.4% lower than that of the national average. The carbon emission in Northwest China in the battery use phase is 28.5 kg CO_2-eq/kWh, which is 81.5% lower than the average level in China. The carbon emission of EVs in the use phase is much lower than that of the conventional internal combustion engine vehicles, even in east China, which has better environmental friendliness. Moreover, with the increasing proportion of renewable energy generation in the electricity mix in China, the battery use phase could achieve near-zero carbon emissions in 2060.

3.3. Carbon emission and carbon reduction potential during battery recycling

3.3.1. Carbon emission results

According to the survey data in Section 2, the LCA of battery material recycling and remanufacturing is conducted in this study. Specifically, the carbon emissions of pyrometallurgy, hydrometallurgy, and direct physical recycling methods are calculated, and the results are shown in Fig. 10. It can be seen that the carbon emission of the pyrometallurgical recycling process is 5.11 kg CO_2-eq/kWh, which is the highest carbon emission among the three recycling methods. In the process of pyrometallurgical, high-temperature smelting batteries with a temperature of more than 1000 °C are usually used. The carbon emission comes from the fossil energy consumed in the metallurgical process, which can produce many direct and indirect carbon emissions. In addition, the graphite in the anode cannot be recycled by pyrometallurgy, and the pyrolysis of graphite in a high-temperature environment will produce carbon emissions. Although pyrometallurgy can bring economic value, its ecological value is not ideal. The carbon emission of hydrometallurgy is 2.68 kg CO_2-eq/kWh, which is 47.6% lower than that of pyrometallurgy. Compared with pyrometallurgy, hydrometallurgy carries out multi-step chemical treatment under low-temperature conditions, without high energy consumption and high carbon emission processes such as high-temperature treatment, and the graphite in anode will not be converted into carbon dioxide to increase carbon emission. Although hydrometallurgy can realize closed-loop recycling, it usually includes more than 10 major steps, which will generate a considerable number of toxic gases and waste solutions (Chen, J. et al., 2019). The carbon emission of the direct physical recycling method is 3.65 kg CO_2-eq/kWh, which is 28.6% lower than that of the pyrometallurgical recycling method, and 26.6% higher than that of the hydrometallurgical recycling method. Since the products of direct physical recycling are materials that can be directly used in battery production, the complicated material reproduction steps and secondary pollution are reduced. The nickel-rich material used in the repair process, such as $Ni_{0.83}Mn_{0.09}Co_{0.08}(OH)_2$ or $Ni(OH)_2$, $LiOH–Li_2SO_4$ in the heat treatment process, and energy consumption are the main sources of carbon emission.

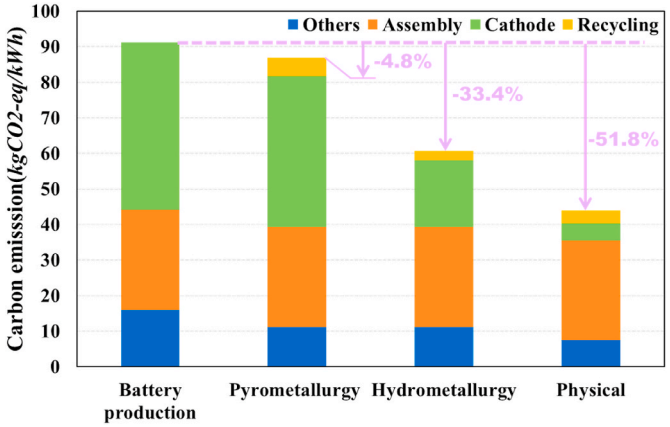


Fig. 10. Carbon emissions of battery recycling and remanufacturing.

3.3.2. Carbon reduction potential of battery recycling and remanufacturing

In this study, the carbon emission reduction potential in the recycling stage is evaluated by comparing the carbon emission of batteries made of recycled materials with that of batteries made of raw materials. Although the recycling process of battery materials will produce carbon emissions, the recycled materials can be directly used to manufacture batteries, avoiding the carbon emissions caused by the mining and refining of raw materials. Fig. 10 shows that the total carbon emission of remanufacturing batteries with recycled materials by pyrometallurgy is 86.86 kg CO_2-eq/kWh, which is 4.8% lower than that of batteries produced with raw materials. It can be seen that pyrometallurgy has a relatively limited carbon reduction potential for LIB remanufacturing. The carbon emission from the remanufacturing of LIBs using the recycled materials from hydrometallurgical is 60.77 kg CO_2-eq/kWh, which is 33.47% lower than that of batteries produced using raw materials. It is because the carbon emission of hydrometallurgy is much lower than that of pyrometallurgy. Moreover, the recycling rate of hydrometallurgy is higher than that of pyrometallurgy (Zhuang et al., 2019). The carbon emission of remanufactured batteries using direct physical recycling is 43.92 kg CO_2-eq/kWh, which is 49.4% lower than that of pyrometallurgy and 27.7% lower than that of hydrometallurgy. It shows the highest carbon emission reduction potential of the three recycling methods. More importantly, the carbon emission of the production of batteries using recycled materials from direct physical recycling is 51.8% lower than that using raw materials. However, the physical recycling method is not mature in technology and is still in the stage of small-scale experiments. Developing efficient and mature physical recycling methods for large-scale applications is critical to reducing carbon emissions.

3.4. Carbon neutrality forecast for battery production

Based on the above discussion, it can be seen that only relying on the recycling of battery materials cannot achieve net-zero carbon emission of battery production in China, and the upgrading of the electricity mix

is a crucial factor. Toward China's carbon neutralization strategy, the carbon emissions in the battery production stage are examined under different electricity mixes in this section, and the results are shown in Fig. 11. It can be observed that the carbon emission of battery remanufacturing with recycled materials under the electricity mix structure in 2030 will be 53.6 kg CO_2-eq/kWh, which is 11.8% lower than that in 2020. The reason for the slight reduction is that thermal power generation still accounts for 50% of the electricity mix in China in 2030. In 2040, the carbon emission of battery remanufacturing with recycled materials will be 38.5 kg CO_2-eq/kWh, which is 36.7% lower than that in 2020. In the same calculation process, the carbon emission from battery remanufacturing will be 9.2 kg CO_2-eq/kWh in 2050, which is 84.9% lower than that in 2020. Combined with battery recycling, CCUS, DAC, and other negative carbon technologies, it is estimated to achieve the carbon neutrality target of the battery industry before 2060. In the short term, the carbon emission of battery production can reduce through the remanufacturing of batteries using recycled materials. In the long term, the carbon neutrality of the battery industry can be achieved by the green electricity mix and negative carbon technologies.

4. Conclusions

In this study, the cradle-to-cradle LCA of the LIBs for EVs in China is conducted, and some valuable carbon reduction measures are given. Moreover, the carbon emission of LIBs production under different electricity mixes is examined in the next 30 years for China's carbon neutral strategy. The main conclusions of this study can be summarized as follows:

(1) Under the current electricity mix dominated by thermal power in China, the carbon emission of the NCM811 battery in the production stage is 91.21 kg CO_2-eq/kWh, in which the production of cathode contributes the most, followed by battery assembly. The carbon emission during battery usage is 154.1 kg CO_2-eq/kWh, mainly from the indirect carbon emission generated by the electricity consumed by the battery. The carbon emission of battery recycling by hydrometallurgy is 2.68 kg CO_2-eq/kWh, which is an ideal recycling method at present.

(2) Increasing the green proportion in the electricity mix and improving production efficiency can reduce carbon emissions. Under China's electricity mix in 2050, the carbon emissions in the production and use stages of batteries will be reduced by 74.8% and 75.1%, respectively. When the battery production efficiency is increased from 30 ppm to 50 ppm, the carbon emission of battery production can be reduced by about 40%. The remanufactured batteries using recycled materials can reduce carbon by 51.8%. In short, green electricity, improving production efficiency, and strengthening battery recycling are necessary carbon reduction measures.

(3) The carbon emission reduction rate of battery production in China will accelerate after the carbon peak in 2030. By 2050, the carbon emission of battery production will be 84.9% lower than that of battery production in 2020. It is predicted that carbon neutralization in LIB production can be realized by optimizing the electricity mix and developing CCUs, DAC, V2G, and other technologies before 2060.

In future work, battery ecological indicators and battery technology factors will be coupled to investigate the sustainability of the battery. For example, the nonlinear ageing of the battery will be considered in the calculation of carbon emissions during battery use.

CRediT authorship contribution statement

Quanwei Chen: Software, Validation, Investigation. **Xin Lai:** Conceptualization, Methodology, Software, Writing – original draft. **Huanghui Gu:** Data curation, Visualization. **Xiaopeng Tang:** Methodology. **Furong Gao:** Conceptualization. **Xuebing Han:** Methodology. **Yuejiu Zheng:** Writing – review & editing.

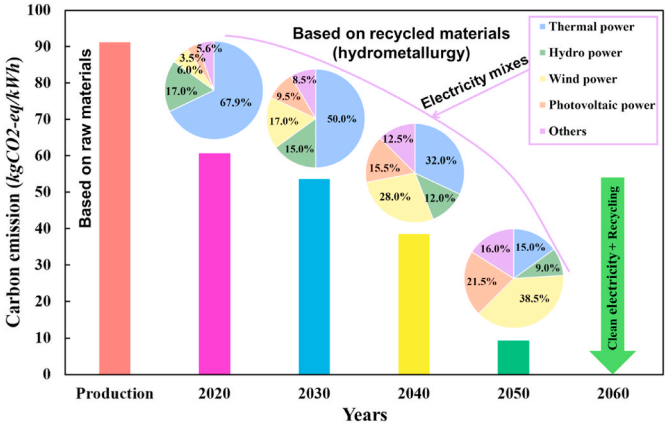


Fig. 11. Carbon emission forecast of battery production.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2022.133342.

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